

AVINASH NANDYALA

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Forward Deployed Engineer for AI voice / conversational platforms - low-latency backend APIs, distributed microservices, and production LLM systems. 3 years at Mastercard shipping high-throughput payment services; MS CS (UMass). Mountain View, CA. F-1 OPT (EAD) - ready to start immediately.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, C#, SQL
Backend & APIs: REST, microservices, FastAPI, ASP.NET, Spring Boot, PostgreSQL
Distributed systems: p95 / latency SLOs, throughput under peak load, idempotent APIs, CI/CD
LLM systems: LLM APIs, structured outputs, RLHF / evals, LangGraph-style agent orchestration
DevOps: Docker, Jenkins, GCP, Azure, Linux, Git

EXPERIENCE

Mastercard Jul 2021 - Jul 2024

Software Engineer 2 Pune, India

- Built distributed Enhanced Currency Conversion (ECC) services (REST / ASP.NET) coordinating FX lookup, conversion, and settlement across gateway microservices; owned conversion-path **p95 latency**, error rate, and throughput under peak international traffic - cut API latency ~20%.
- Shipped PCI key-block compliance (CAVV, HMAC) on tokenization microservices with 100% test pass on fraud-sensitive paths - production reliability for security-critical APIs.
- Led Java 8 → Zulu 17 modernization across gateway / tokenization services; fixed Jenkins CI failures and remediated production CVEs to keep customer traffic stable.
- Partnered with security and infrastructure teams on PCI-aware payment platforms used at global scale - strong fit for customer-facing FDE delivery.

Biomedical NLP Lab, UMass Amherst Feb 2025 - May 2025

Graduate Research Assistant Amherst, MA

- Production-minded LLM work: fine-tuned / aligned Qwen & LLaMA with instruction tuning + RLHF for clinical ranking; ~85% accuracy with RLHF vs ~60% SFT; 27 w-F1@Recall10 on MIMIC-EXT-IV / RARE.
- Built evaluation harnesses for LLM quality (ranking, coherence) - transferable to voice-agent quality, latency, and regression checks.
- **Hierarchical Context Aggregation for Medical Notes** | [GitHub](#): spoken-language normalization and hierarchical clustering for automated SOAP notes from doctor-patient transcripts.
- Lowest GPT-2 perplexity (21.35 vs 26.51 baseline) and BioBERT embedding similarity 0.3148 among clustering approaches on 207 ACI-Bench transcripts.

PROJECTS

FleetControl | App | [GitHub](#) | *React, TypeScript, agent graphs* 2026

- Ops control plane syncing status, gates, spend, and CI across 10+ services - the kind of multi-system integration FDEs ship with customers.
- Built LangGraph-style stateful agent orchestration in-house (graph nodes, tools, verification gates, provenance) for automated fleet workflows.

ComicMainEngine | App | [GitHub](#) | *FastAPI, LLM APIs* 2026

- End-to-end LLM product API: FastAPI routes text models + image APIs, with vision quality gates and automatic retries before output ships.
- Usage accounting and failure recovery - production patterns for reliable generative / conversational backends.

Procedural City | [GitHub](#) | *Python, structured LLM outputs* 2026

- LLM Scene Director emits schema-validated structured outputs that drive a 3D pipeline; hard validation so bad generations never reach runtime.

EDUCATION

University of Massachusetts Amherst Amherst, MA

M.S., Computer Science May 2026

Indian Institute of Technology (IIT) Kharagpur Kharagpur, India

B.Tech., Computer Science and Engineering May 2021

HONORS

IIT Kharagpur - JEE Advanced (2017): Top 0.1% of 1.2M+ applicants.